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John B. Schlaerth III, M.S.

WORK EXPERIENCE

Lead Software/Hardware Engineer, Agentic AI IRAD Initiatives – L3Harris LHX Accelerator Melbourne, FL: *10/25 – Present*
- 4th place out of 73 competing teams, spearheading the Tactical Field Agent: An edge AI assistant, manpack radio platform MVP
- Finalist, creating an autonomous ground robot platform architecture, in collaboration with startup partner Polymath Robotics

Lead Project/Component Engineer, LDACS IRAD Program – AXYV/L3Harris Missile Solutions Los Angeles, CA: *2/26 - Present*
- Led Combustion Devices IPT for DACS IRAD program, spearheading hotfire & electronic valve testing, planning, & procurement

Member, Engineering Leadership Development Program (ELDP) – L3Harris CHQ Melbourne, FL: *6/24 – 6/26*
- Selected for accelerated rotational program focused on advanced defense systems, technical leadership, & AI integration

Electrical Design Engineer, USAF Big Safari Program – L3Harris Space & Mission Systems Greenville, TX: *6/25 – 2/26*
- Designed military aircraft electrical diagrams & led HDL harness OPs coordination, modernizing C-37 & VC-25 electronic capabilities
- Created error-filtering Artificial Intelligence process flow adopted by program management for use in standard design operations

FPGA Digital Design & Software Engineer, Black Side Digital Design – L3Harris Comm. Systems Salt Lake City, UT: *6/24 – 6/25*
- Designed & validated VHDL RTL circuits w/ self-checking testbenches for black-side TX/RX modules on FPGAs for SDA DVB satcom
- Analyzed electrical reliability & safety of 200+ embedded components for RAMS group in broadband radio integrated circuits
- Developed autonomous Python crawler bot, using AI to accelerate website migration: Eliminated 1,000+ hrs manual upgrade work
- Created comprehensive FPGA training course for L3Harris employees, covering full CS-W FPGA development and build cycle

Lead Project/Product Engineer, Next Generation Interceptor – Aerojet Rocketdyne/L3Harris Los Angeles, CA: *6/23 - 6/24*
- Managed engineering team of 8 to coordinate design, testing, and on-time delivery of advanced hypergolic rocket thrusters
- Led a 5-person trim-test team to clear a critical-path milestone; owned manifold hardware delivery and drove CDR with ~30 stakeholders; stood up a 4-person metal additive manufacturing strikeforce that cut component lead time by 16 weeks
- Analyzed >200 PAC-3 MSE solid rocket motor propellant X-ray scans, classifying 500+ void anomalies for evaluation & analysis

LEADERSHIP, PROJECTS & RESEARCH

Electronics/Project Engineer, FB-V12/HM-V2 Robot Design Project – Columbia University New York City, NY: *1/23-5/23*
- Invented 2 unique walking robots, built with generative optimization, 3D printing, and Python-coded control/test algorithms

Software Engineer, Simulated Soft Robotics Project – Columbia University New York City, NY: *8/22-12/22*
- Coded advanced evolutionary algorithms AI from scratch in C++ and Python, evolving walking soft robotic designs

Team Lead, Atmospheric Doppler LIDAR Scanner (ADLS-V9) Project – NOAA Boulder, CO: *8/21-5/22*
- Spearheaded engineering team of 9 with $50,000 budget to construct robust and ultra-high precision LIDAR scanner for NOAA

Team Lead, Recreational Vehicle Engineering Redesign Project – University of Colorado Boulder, CO: *1/22-5/22*
- Directed team of 5 to develop mechanical recreational vehicle prototypes and systems for the elderly and disabled

Design Engineer, Moving Highway Kinetic Sculpture Project – University of Colorado Boulder, CO: *1/22-5/22*
- Devised and manufactured a high-precision electromechanical kinetic sculpture engineering project with team of 7 ME students

Team Lead, Air-Form Splint Biomedical Engineering Manufacturing Project – University of Colorado Boulder, CO: *1/21-5/21*
- Guided engineering team of 4 in developing emergency medical splint product, conducting testing and analysis on prototypes

Design Engineer, Anti-Nuclear Blast Shelter Research and Design Project – University of Colorado Boulder, CO: *1/21-5/21*
- Designed and analyzed shelter system capable of extreme nuclear blast resistance, withstanding high yield blast from 1 km away

Design Engineer, Reverse Engineering CAD Modeling Project – University of Colorado Boulder, CO: *1/20-5/20*
- Performed intricate reverse engineering on an industrial battery charging apparatus and created detailed 3D SolidWorks model

Team Lead, Electronic Bike Lock Engineering Design Project – University of Colorado Boulder, CO: *8/19-12/19*
- Led team of 5 to develop a high-fidelity prototype of an electronic bike lock using Arduino, FDM 3D printing, and CNC milling

SKILLS & CERTIFICATIONS

- **Organization:** US Secret Security Clearance, Jira, Project/Team Management, MS Project, CAM/EVMS, Customer Communication
- **Electrical**: Vivado, Xilinx SoCs, Siemens NX, Capital Logic/Harness XC, LTspice, KiCad, Arduino IDE, Etcher, SBC, Solder/Harness Ops
- **CompSci:** VHDL, Verilog, Matlab, C, C++, Python, Javascript, GitHub/Bitbucket, Git, SQL, Linux, Bash, Docker, OOP/FP, MCP
- **Artificial Intelligence (AI):** DNN/CNNs, RL, Agents, PyTorch, TensorFlow, Machine Vision, Edge/Embedded AI, EA/GAs, Local LLMs
- **Manufacturing:** G-code, Saws, Drills, Mills, Lathes, FDM/SLM Printing, EDM, MIG Welding, Brazing, Laser Cutting, CNC, Casting
- **Software/CAD:** SolidWorks, Creo, Onshape, Blender, Altair, Photoshop, Fusion360, Autocad, Finite Element Analysis, GUI Design
- **Specialties:** Rocketry, Autonomous Systems, Robotics, Embedded Electronics, Simulation, Product Design, Full-stack SWE, EA CAD

EDUCATION

Columbia University, Master of Science in Mechanical Engineering New York City, New York
Concentration in Robotics and Control Systems August 2022 - May 2023

University of Colorado Boulder, Bachelor of Science in Mechanical Engineering Boulder, Colorado
Concentration in Advanced Product Design and Entrepreneurship August 2018 - May 2022